
Wildfire Smoke Detection from UAV Imagery

Location-based evaluation of a lightweight YOLO11n detector

Dataset

Boreal Forest Fire UAV image subset from Fairdata

Evaluation completed July 28, 2026

Videos excluded; labeled UAV images only

Executive findings

4,954	Labeled images
4,693	Smoke-positive images
Heinola	Hardest held-out location
320x180	Lowest tested reliable resolution

1. Dataset inventory

The labeled subset contains image and YOLO label folders for Evo, Heinola, Karkkila, and Ruokolahti, plus Empty-Images and Empty-Labels. Each row stores normalized class, x-center, y-center, width, and height. The only class is 0: smoke.

Location	Images	Smoke	No smoke	Boxes
Evo	931	930	1	957
Heinola	1,123	906	217	917
Karkkila	1,133	1,095	38	1,104
Ruokolahti	1,767	1,762	5	1,884
Total	4,954	4,693	261	4,862

Ground-truth examples



Yellow rectangles are the supplied smoke bounding boxes. The audit found 4,862 boxes across 4,693 smoke-positive frames.

Original image dimensions: 4096x2160 (3,728 images), 3840x2160 (1,133), and 1920x1080 (93).

2. Experimental design

Four leave-one-location-out models were trained. In every fold, three complete forest locations supplied training data and the fourth remained unseen until testing. A stratified 10% validation subset was drawn only from the three training locations.

Held-out	Train	Validation	Test	Training locations
Evo	3,620	403	931	Heinola, Karkkila, Ruokolahti
Heinola	3,448	383	1,123	Evo, Karkkila, Ruokolahti
Karkkila	3,439	382	1,133	Evo, Heinola, Ruokolahti
Ruokolahti	2,867	320	1,767	Evo, Heinola, Karkkila

Training configuration

Setting	Value
Model	YOLO11n, approximately 2.58 million fused parameters
Initialization	Pretrained yolo11n.pt weights
Epochs	15
Training input	640 pixels
Batch size	8
Optimization	Automatic optimizer, cosine schedule, mixed precision
Reproducibility	Deterministic seed 20260728
Hardware	NVIDIA GeForce RTX 4060, 8 GB

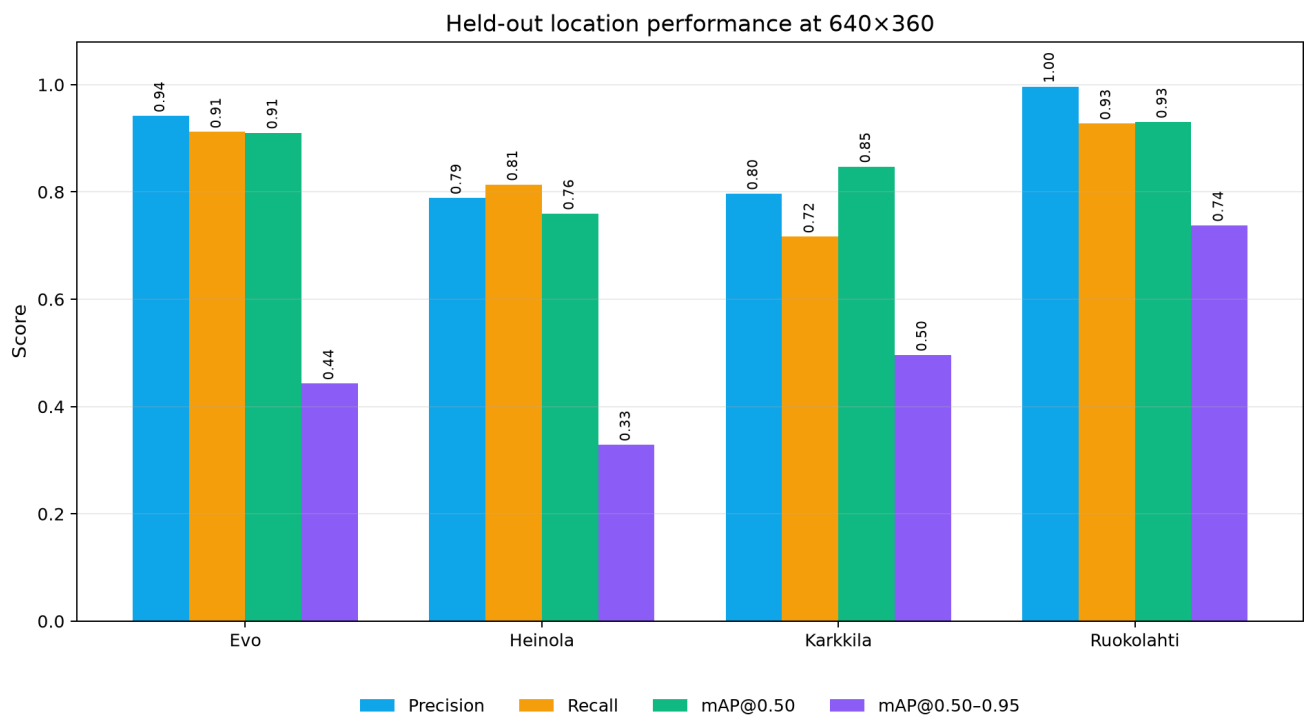
Evaluation metrics

Precision measures how many reported boxes are correct. Recall measures how many ground-truth boxes were found. mAP@0.50 accepts matches with at least 0.50 intersection-over-union. mAP@0.50-0.95 averages increasingly strict thresholds and emphasizes accurate plume localization.

3. Cross-location performance

The 640x360 setting is most directly comparable to the 640-pixel training scale. Each row comes from a different model whose named test location was never present in training or validation.

Held-out	Precision	Recall	mAP@0.50	mAP@0.50-0.95
Evo	94.2%	91.2%	91.0%	44.3%
Heinola	78.9%	81.4%	76.0%	32.9%
Karkkila	79.7%	71.6%	84.7%	49.6%
Ruokolahti	99.7%	92.8%	93.0%	73.8%



mAP@0.50-0.95 is lower because smoke plumes have diffuse boundaries and strict box overlap is difficult.

Hardest overall: Heinola. It had the lowest mAP values at 640x360 and the lowest mean mAP@0.50-0.95 across all four resolutions. Karkkila had the lowest recall at 640x360.

4. Resolution comparison

Exact source copies were generated at 1920x1080, 1280x720, 640x360, and 320x180. Each copy was evaluated using the corresponding inference canvas size. Values are macro averages over the four held-out-location folds.

Resolution	Precision	Recall	mAP@0.50	mAP@0.50-0.95	Worst recall
1920x1080	17.1%	17.5%	8.3%	1.9%	0.6%
1280x720	55.4%	56.7%	50.6%	17.7%	28.2%
640x360	88.1%	84.2%	86.2%	50.1%	71.6%
320x180	88.3%	86.2%	87.1%	51.5%	77.1%



The dashed series is the macro mean across the four held-out locations.

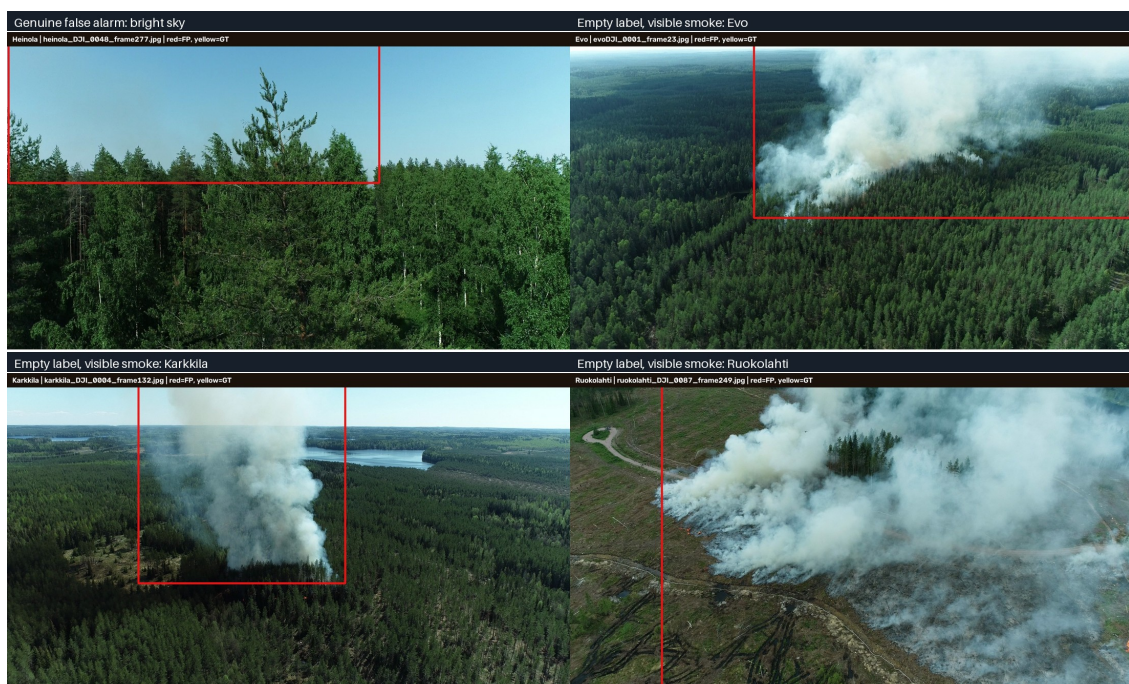
Lowest tested reliable resolution: 320x180. Reliability required macro recall $\geq 80\%$, macro mAP@0.50 $\geq 80\%$, and recall $\geq 70\%$ at every held-out location.

Important interpretation: the network was trained at a 640-pixel canvas. Direct 1280- and 1920-pixel inference produced a severe object-scale mismatch. This does not mean high-resolution cameras are intrinsically worse; deployment should resize large frames to the trained scale or retrain with multi-scale inputs.

5. Incorrect detections and label quality

Errors were audited at 320x180 using confidence ≥ 0.25 and intersection-over-union ≥ 0.50 . False-positive boxes include duplicate or oversized plume boxes in smoke-positive frames, so they do not always represent confusion with a different object.

Location	FP images	FP boxes	Missed boxes	No-smoke	No-smoke flagged
Evo	59	64	77	1	1
Heinola	252	254	282	217	1
Karkkila	274	314	130	38	2
Ruokolahti	7	7	143	5	3



Top left: genuine bright-sky/haze false alarm. The other examples have empty labels but visibly contain smoke.

Seven nominal no-smoke frames were flagged. Visual review found obvious smoke in six despite empty label files. The one genuine environmental false alarm was **heinola_DJI_0048_frame277.jpg**, where a broad box covered bright, slightly hazy sky above treetops. No clear cloud-only, fog-only, or water-reflection false alarm was observed at the selected threshold.

6. Conclusions and deliverables

- A compact YOLO11n detector generalized to unseen forest locations when inference scale aligned with training.
- At 640x360, held-out mAP@0.50 ranged from 76.0% in Heinola to 93.0% in Ruokolahti.
- 320x180 remained reliable for smoke-presence detection under the stated rule, although precise plume boundaries remained harder.
- Heinola was hardest overall, while Karkkila had the weakest 640x360 recall.
- Label omissions materially affected false-positive interpretation: six of seven flagged nominal backgrounds visibly contained smoke.
- For deployment, resize large UAV frames to the trained scale or retrain with multi-scale inputs before direct high-resolution inference.

GitHub submission contents

Folder	Contents
data	All 4,954 labels, portable splits, index, counts, and official archive hashes
models	Best checkpoint from every leave-one-location-out fold
results	All metrics, per-image errors, figures, curves, and sanitized run settings
scripts	Audit, preparation, training, resolution evaluation, and error analysis
docs	This PDF report

Source and reproducibility

Dataset: etsin.fairdata.fi/dataset/1dce1023-493a-4d63-a906-f2a44f831898

DOI: [10.23729/fd-72c6cf74-b8eb-3687-860d-bf93a1ab94c9](https://doi.org/10.23729/fd-72c6cf74-b8eb-3687-860d-bf93a1ab94c9)

Dataset license: CC BY 4.0

All numerical values in this report come from the saved audit files and 16 measured held-out evaluations included in the repository. Video files were not analyzed. Raw image binaries remain in the official Fairdata record rather than GitHub.